

Software Requirements Specification (SRS)

Centralized Marine & Automation E-commerce Helpdesk

1. Introduction and Project Scope

1.1. Project Purpose

The objective of this project is the development of a centralized, multi-tenant helpdesk platform engineered specifically for the marine and automation industrial sectors. The system serves as a unified communications hub, aggregating disparate data streams from five specific e-commerce environments into a single-pane-of-glass interface. This consolidation is designed to optimize support workflow, reduce response latency, and provide high-fidelity context for technical sales and service inquiries.

1.2. Platform Scope

The platform architecture provides native integration support exclusively for the following channels:

- eBay
- Shopify
- auto2mation
- aeliyamarine
- aeliyamarinetech

And there may be chance that in future we add another ebay or mail id so we need to make it scalable. So the system will get designed such that the operation head will get an option to add the account id and link it with the ebay or gmail so the other details will be also asked like logo image, platform name and any other platform details.

Exclusion Criteria: To maintain the specialized industrial focus of the tool, integration with generic consumer-facing platforms such as Amazon or Flipkart is explicitly excluded from the current scope.

1.3. Target Audience and Tone

This specification is intended for Technical Product Managers, System Architects, and Customer Support Lead Developers. The tone is strictly technical, structured, and geared toward system implementation and architectural integrity.

2. Functional Requirements

2.1. Centralized Inbox & Multi-Channel Management

- **Unified Interface:** The UI must aggregate all incoming tickets into a singular stream. Every ticket entry shall feature high-visibility platform-specific logos to facilitate immediate source identification.
- **Multi-Tenant Architecture:** The dashboard must accommodate multi-tenancy, allowing a single authenticated user session to manage multiple distinct accounts per platform (e.g., three separate eBay storefronts) with strict logical separation.

2.2. AI-Driven Message Classification

The system shall implement an automated classification engine to tag incoming communications based on intent. Required categories include:

- Cancellation
- Refund
- Order Update
- Return request
- Pre-sales
- Irrelevant/Spam: Required to filter noise from the primary workflow.

2.3. SLA Management & Response Prioritization

- **SLA Parameters:** The system shall enforce a 24-hour response target from the timestamp of ticket creation.
- **"Respond Till" Visual Progress Bar:** Every ticket entry must display a visual countdown labeled "Respond Till." This indicator shall be a receding horizontal progress bar.
- **Color-Coded Urgency Logic:**
 - Green: >18 hours remaining; bar is full.
 - Yellow/Orange: 6–17 hours remaining; bar is partially receded.
 - Red: <6 hours remaining; bar is nearly empty, indicating critical urgency.

2.4. Synchronization & Integration Logic

- **Inbound Data Flow:** The system shall utilize Webhooks for real-time message ingestion to ensure sub-second latency from the source platform to the helpdesk.
- **Outbound Data Flow:** Replies must be pushed via REST API endpoints to the respective platform.
- **Dynamic Templating:** Agents shall have access to response templates utilizing liquid-style syntax to inject customer names, order IDs, and shipping statuses.

2.5. Contextual Information Sidebar

A real-time data sidebar shall activate upon ticket selection, providing:

- Customer Metadata: Display name, country of origin (via flag icons), and a VIP tag for high-value or frequent buyers.
- Order Intelligence: Current Sales Order (S/O) status and active tracking links.
- Sales Value: Total Sales Order value, defined as a currency-formatted decimal (e.g., \$933.30) supporting multiple ISO currency symbols.

2.6. Agent Collaboration & Workflow

- Collision Detection: To prevent redundant responses, the "Owner" column must update in real-time with agent avatars or initials (e.g., "DJ", "DB") when a ticket is being viewed or edited.
- Internal Communication: Support for threaded internal comments with @mention functionality for team escalation.

3. Technical Architecture & System Requirements

3.1. Technology Stack

Layer	Technology
Frontend	React TS (TypeScript)
Backend	Python FastAPI
Database	PostgreSQL (Alembic for migrations)

Rationale for FastAPI: Selected for its asynchronous capabilities, which are critical for handling high-concurrency API polling and concurrent webhook processing without blocking the event loop.

3.2. Database Schema Design

The system shall implement Row-Level Security (RLS) within PostgreSQL to maintain strict multi-tenant isolation. This ensures that while all messages reside in a centralized repository for the AI engine, data access is restricted at the database level based on the tenant_id associated with the specific platform account.

4. UI/UX Design Specifications

4.1. Dashboard Layout & Column Configuration

The main "Tickets" or "To do" view shall adhere to the following column-based structure:

1. Source: Icon-based column representing the originating platform (e.g., eBay logo).
2. Account: Text-based label identifying the specific store instance.

3. Customer: Includes customer name, country flag, and "VIP" status tag.
4. AI Classification: Tag-based display (e.g., "Pre-sales").
5. Subject: Latest message snippet or subject line.
6. Owner: Visual indicator of assignment or collision (Agent initials/avatar).
7. Respond Till: The receding color-coded SLA progress bar.
8. S/O Value: Currency-formatted decimal value associated with the inquiry.

4.2. Global Navigation & Ticket States

The sidebar and top-level UI must provide the following navigation and control states:

- Primary States: Dashboard, Tickets, Chats, Calls.
- Ticket Filters: All Tickets, My Tickets, Unassigned, Starred, Resolved, Not Resolved, Snoozed, Delayed, Blocked, Archive.
- Global Controls: A global "Search" bar and a "Filters" button located in the top-tier header for rapid ticket retrieval.

5. Analytics and Performance Tracking

5.1. Key Performance Indicators (KPIs)

The platform shall track and visualize the following metrics:

- Average Response Time (ART): The mean time elapsed from the initial ticket timestamp to the first outbound agent reply.
- Volume Trends: A comparison of "Tickets Created" vs. "Tickets Resolved" over a sliding 7-day and 30-day window.
- Granular Filtering: All KPI reporting must be filterable by the five supported channels (eBay, Shopify, auto2mation, aeliyamarine, aeliyamarinetech) and by individual agent accounts.